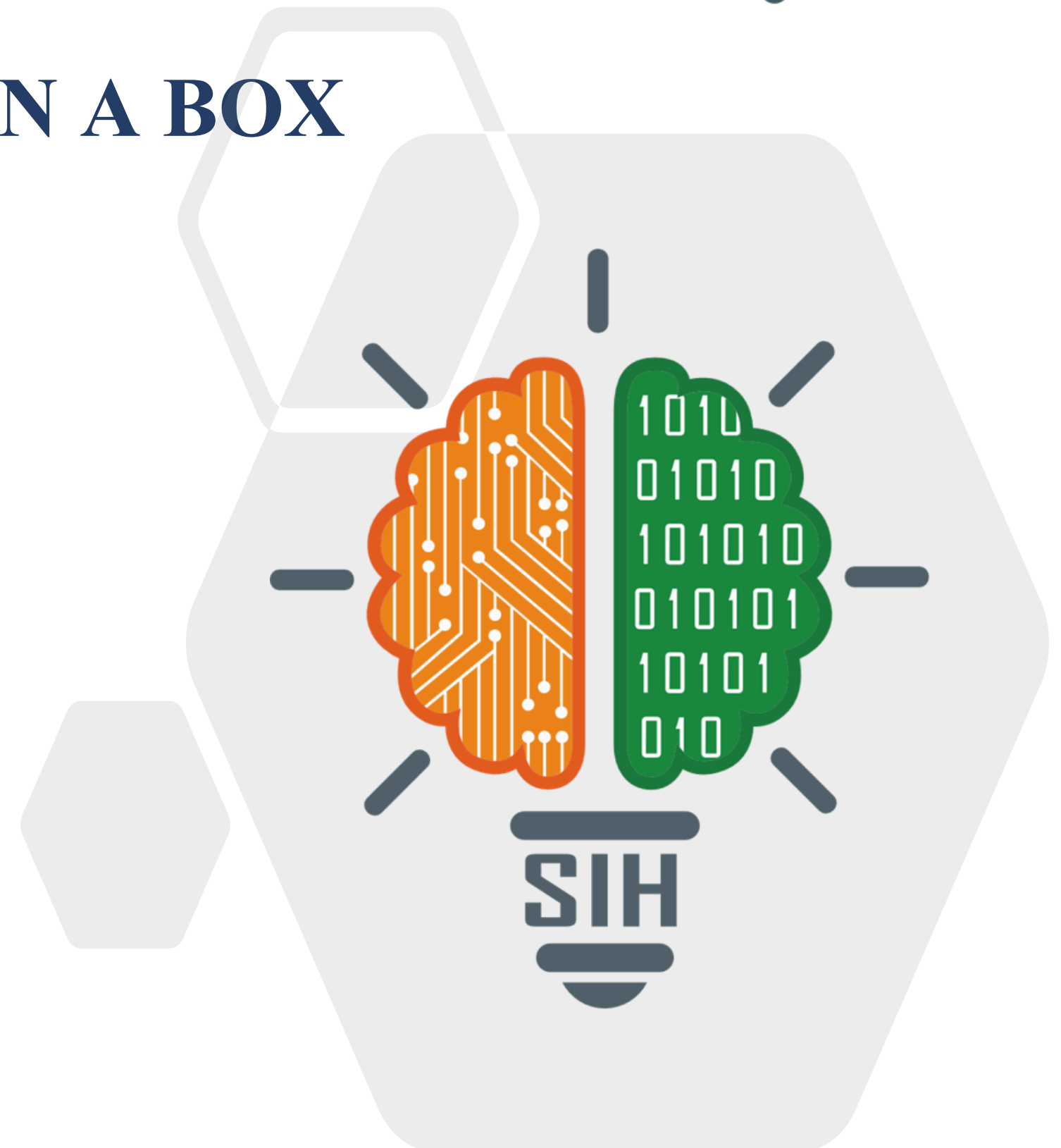


GRAM-JARVIS : PHC IN A BOX

- **Problem Statement ID** – SIH26133
- **Problem Statement Title** – Accessibility and quality of public healthcare services, particularly in rural and underserved areas
- **Theme** – MedTech/BioTech/HealthTech
- **PS Category** – Software
- **Team ID** – 141
- **Team Name** – INVINCIBLE_RGUKTN



Voice-first family app + offline PHC Edge Hub — strengthens ASHA & PHC, never replaces them

PROPOSED SOLUTION

SOFTWARE • FAMILY BOT NETWORK

One Family ID → 5 bots: Father • Mother • Child • Pregnancy • Elder
Voice-first in Telugu / Hindi / Tamil — zero typing

HARDWARE • PHC EDGE HUB (USP)

Raspberry Pi + battery + WiFi + thermal printer
Works offline • syncs at night • prints token & slips

ASHA-ASSIST MODE

Simple 3-button app • works offline
ASHA connects patient to PHC doctor, even on 2G

HOW IT ADDRESSES THE PROBLEM

- 01

Distance & queue

Smart token + live medicine/X-ray status — check before travelling.
- 02

Language & literacy

Bhashini voice in/out. Villager speaks, hears and reads printed Telugu.
- 03

Fragmented records

ABHA-linked longitudinal record stored in hub, synced via FHIR/ABDM.
- 04

Broken referral

QR referral slip — district hospital scans it, completion is tracked.
- 05

No follow-up

Voice reminders for ANC/BP; Red Alert to doctor + 108 for high risk.



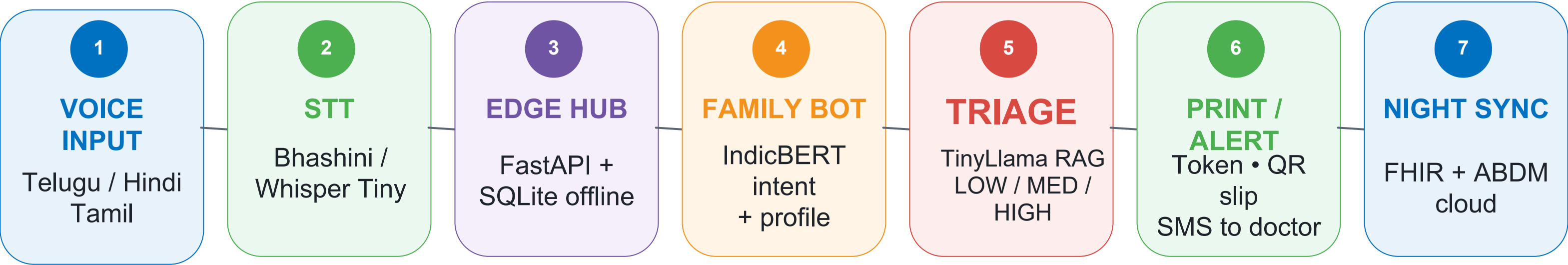
PHC EDGE HUB
One box per PHC

WHY UNIQUE

Offline-first • Voice + Print (paper builds trust) • Family bots • Works WITH ASHA/PHC via ABDM/FHIR

Offline-first pipeline: voice input → PHC Edge Hub → triage → print / alert → night sync to ABDM cloud

Flutter • FastAPI • SQLite / PostgreSQL • Whisper + Bhashini • TinyLlama RAG • FHIR R4 / ABDM • Raspberry Pi 4



RED — HIGH RISK

Red Alert: auto SMS/IVR to doctor + 108 • fast escalation

YELLOW — MEDIUM

PHC doctor referral • token print • QR referral slip • follow-up

GREEN — LOW RISK

Home-care advice in local language • scheme alerts • routine tracking



FAMILY BOT APP

Voice → save locally on hub → print slips → night sync → secure FHIR/ABDM record • AES-256 + TLS 1.3

Buildable in ~3 months, low cost, and delivered through existing ASHA / PHC staff

FEASIBILITY

- ✓ **TECHNICAL** Open-source stack; Bhashini is free and ABDM sandbox is available for the prototype.
- ✓ **FINANCIAL** Hub ~₹25K + ₹50K/PHC/year AMC, paid by govt / NGO / CSR.
- ✓ **OPERATIONAL** Uses existing ASHA workers and PHC staff — no new hiring; 3-button ASHA mode.
- ✓ **SCALABLE** Modular low-cost parts; local technicians can replace them; OTA software updates.

CHALLENGE

MITIGATION

Connectivity / power

Offline hub + UPS battery
night sync to cloud

Language / literacy

Bhashini voice-only UI
+ printed Telugu slips

Privacy & consent

ABDM consent + FHIR
AES-256 + role-based access

Sync conflicts

Timestamps + conflict rules
/ CRDT-style resolution

Adoption & upkeep

ASHA-led onboarding, pictorial UI
modular parts, OTA updates

MODEL (B2G2C): 100 PHCs ≈ ₹50 lakh / year recurring • MONITOR → MITIGATE → SECURE → ADAPT

TARGET AUDIENCE

- Patients: less travel, records follow them
- ASHA: less paperwork, incentive tracking
- PHC / Govt: live OPD, stock, referrals

BENEFITS

- Social: better ANC follow-up, early triage
- Economic: travel cost saved per patient
- Sustainable: govt pays, not the poor farmer
- Unique: offline + voice + family bots + hardware

QUANTIFIABLE TARGETS

- 70% less unnecessary travel
- 60% faster referral completion
- 60% less ASHA paperwork
- 100% record continuity, even offline

IMPACT ECOSYSTEM

Five pillars, one platform — offline, voice-first and family-centric

**EARLY
TRIAGE**

early risk signals

**GRAM-
JARVIS****ASHA
EMPOWERED**

less paperwork

**REFERRAL
TRACKING**

QR closed loop

**MEDICINE
VISIBILITY**

know before you go

**OFFLINE
ACCESS**

rural reach

SOCIAL • ECONOMIC • SYSTEM-LEVEL — strengthens govt, not replaces

ABDM

Identity, consent and record exchange

abdm.gov.in**BHASHINI**

Indian language speech and translation services

bhashini.gov.in**MoHFW / IPHS**

PHC service and infrastructure context

iphs.mohfw.gov.in**HL7 FHIR R4**

Interoperable health record structure

hl7.org/fhir/R4/**eSanjeevani**

Existing public teleconsultation workflow

esanjeevani.mohfw.gov.in**Software prototype :** <https://lovely-sorbet-39c476.netlify.app/>**software explanation :** https://youtu.be/O7ecTP_PzIY?si=UbSn6PmuF594xGBe**Hardware prototype :** <https://spiffy-duckanoo-df90aa.netlify.app/>**Hardware explanation :** <https://youtu.be/l7j7x8-ERS0?si=rRbgyFc6l4cpWKq7>